AI-Integrated Washing Machine

Mini Project Presentation

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Outline

1. Introduction
2. Hardware Design
3. AI & Backend
4. Mobile App & Testing
5. Results & Future Scope

Introduction

Problem Statement & Objective

- **Problem:** Existing semi-automatic machines waste water, detergent, and often damage clothes due to fixed wash cycles.
- **Objective:** Convert a semi-automatic washing machine into a fully automated, AI-powered system that:
  - Identifies cloth type (fabric, fiber category, soil level) using a smartphone camera.
  - Optimizes wash parameters (detergent, water level, spin time, temperature, mechanical action) to extend cloth lifetime.

Goal: *Intelligent washing that preserves fabric quality and conserves resources.*

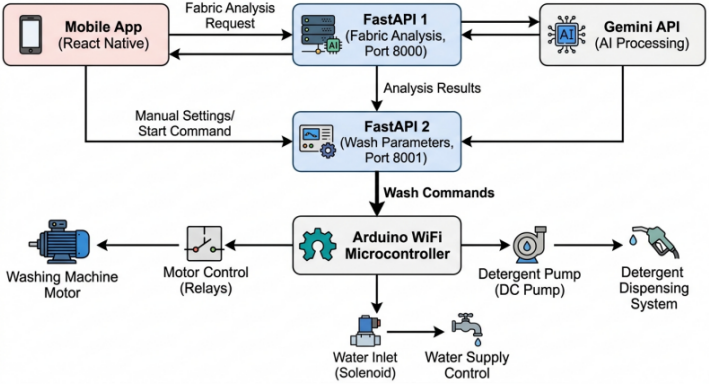
System Architecture

Cloth Image $\rightarrow$ **Mobile App** $\rightarrow$ **Gemini API** $\rightarrow$ **Arduino** $\rightarrow$ **Washing Machine**

Two-Stage AI Pipeline:

1. **Fabric Detection** – Identifies fabric type, fiber category, and soil level.
2. **Wash Parameter Calculation** – Computes optimal wash settings based on Stage 1 output.

System Architecture Overview



Flow: Image → Gemini (Stage 1 & 2) → Arduino → Machine Actuation

Hardware Design

Hardware Components

Core & Commutation

- Semi-automatic Washing Machine
- Arduino WiFi R4
- 3-Phase Motor (240V, 10A)
- 2 Rotation Relays + 1 Solenoid Relay

Dispensing & Actuation

- Solenoid Water Inlet Valve
- 5V DC Water Pump (Detergent)
- Transistor TAS122
- 12V L298N DC Motor Driver

All components integrated to automate washing, dispensing, and motor control.

Hardware Design & Control Logic

- **Motor Control:** Clockwise 5s $\rightarrow$ Stop 2s $\rightarrow$ Counter-clockwise 5s (loop) using relays.
- **Detergent Dispenser:** Linear regression model calculates valve open time.
  - Data collected: volume dispensed at 50 ms, 100 ms, . . . , 2400 ms intervals.
  - Linear regression equation implemented in Arduino code for precise dispensing.
- **Water Inlet Valve:** Same calibration approach ensures accurate water filling.

Linear Regression Example

$$\text{Open Time (ms)} = a \times \text{Desired Volume (ml)} + b$$

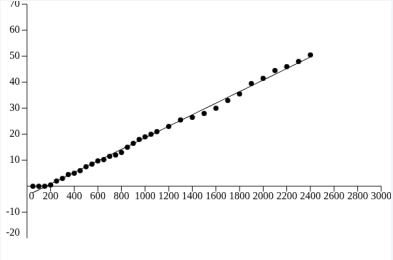
(coefficients a, b obtained from experimental calibration)

Detergent Dispenser Calibration

Calibration Data

Time (ms)	Vol (ml)	Time (ms)	Vol (ml)	Time (ms)	Vol (ml)
50	0.0	650	10.25	1400	26.5
100	0.0	700	11.5	1500	28.0
150	0.0	750	12.0	1600	30.0
200	0.5	800	13.0	1700	33.0
250	2.0	850	15.0	1800	35.5
300	3.0	900	16.5	1900	39.5
350	4.5	950	18.0	2000	41.5
400	5.0	1000	19.0	2100	44.5
450	6.0	1050	20.0	2200	46.0
500	7.5	1100	21.0	2300	48.0
550	8.5	1200	23.0	2400	50.5
600	9.75	1300	25.5	–	–

Linear Regression Model



Coefficient	Value
a (slope)	0.022176
b (intercept)	-3.569687
R^2 value	0.997 (approx.)

AI & Backend

Fabric Analysis (Gemini API)

Multi-stage prompt system extracts:

- Fabric name
- Fiber category
- Soil level (1–5 scale)

Visual markers analyzed:
diagonal ribbing, fiber luster, fold behavior.

Soil Level Scale

Level 1: Clean

Level 3: Moderately Dirty

Level 5: Emergency

JSON Output Format

Output JSON (Stage 1)

```
{
  "material_type": "cotton twill",
  "fiber_category": "Natural",
  "description": "Heavy cotton fabric with diagonal ribs",
  "dirt_level": 3,
  "confidence_score": 0.92
}
```

Wash Parameter Calculation

Takes JSON from Stage 1 and computes optimal wash settings:

- Detergent: baseline 20 ml, adjusted
- Soak: 0-30 min
- Spin: 3-12 min based on fabric
- Water: standard 35 L
- Wash Cycles: (1–2)
- Temp: up to 60 °C
- Action: Gentle / Normal / Heavy

Wash Parameters JSON

Output JSON (Stage 2)

```
{
  "detergent_ml": 28,
  "soak_minutes": 15,
  "spin_minutes": 8,
  "water_liters": 35,
  "wash_cycles": 2,
  "temperature_c": 45,
  "mechanical_action": "Normal"
}
```

FastAPI Backend

Two independent microservices:

API 1 – Fabric Analysis (Port 8000)

- Input: Cloth image
- Uses Pillow + Gemini API
- Predicts: fiber type, dirt level, description
- **Human-in-the-Loop** feedback for correcting predictions

API 2 – Wash Parameter Calculator

Tech Stack: Python, FastAPI, Pydantic, Pillow, Gemini API, OpenAPI

Mobile App & Testing

User Interface – Tab-Based Navigation

5 Screens

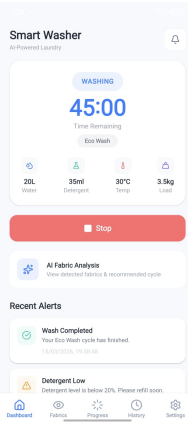
1. Dashboard
2. Fabric Analysis
3. Wash Progress
4. Manual Override
5. Settings

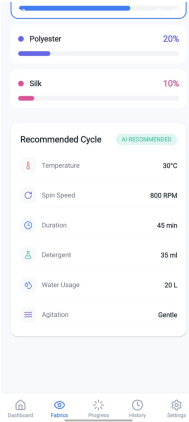
Key Features

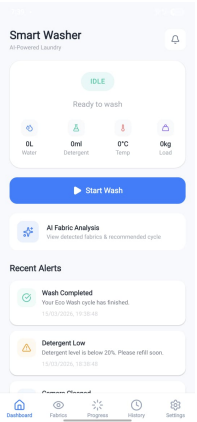
- Reusable UI components (Step Indicators, Notification Cards)
- TypeScript interfaces for data integrity (Fabric, MachineMetrics, CycleState)
- Tailwind CSS utility-first styling
- Async API service layer connecting to backend
- Cross-platform (iOS & Android)

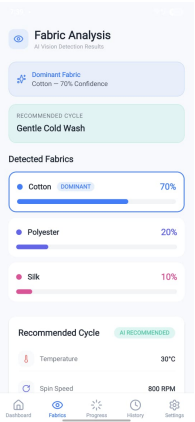
App communicates with FastAPI backends via REST calls.

Mobile App Screenshots (1/2)

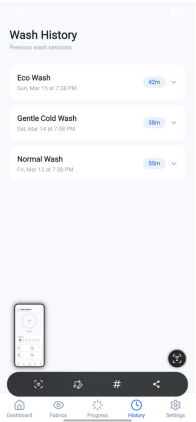


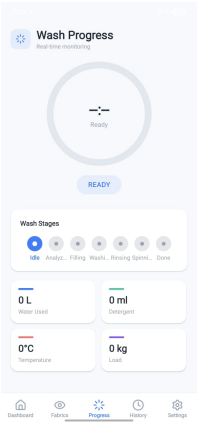


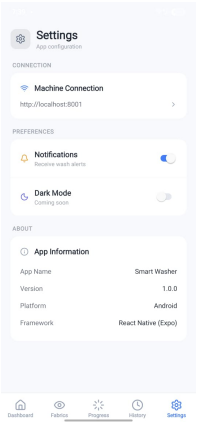




Mobile App Screenshots (2/2)







Dataset Validation & Testing

- Test cases designed from **heavy-duty (Denim)** to **delicate (Chiffon)**.
- **Human-in-the-Loop (HITL)** testing for misclassifications:
  - Example: Differentiating *Cotton Twill* vs *Synthetic Blends*
  - Iterative prompt refinement eliminated identification errors.
- Standardized JSON output ensures seamless hardware integration.
- Validation of detergent dispensing using linear regression (error $< 5\%$).

Result

Reliable classification across 15+ fabric types with high confidence.

Results & Future Scope

Results & Key Achievements

- **Accurate Fabric Classification:** High confidence scores across multiple fabric types.
- **Precise Detergent Dispensing:** Linear regression validated against real measurements.
- **End-to-End Automation:** Image scan $\rightarrow$ AI decision $\rightarrow$ Hardware execution.
- **Increased Cloth Lifetime:** Optimized parameters per fabric type reduce wear and tear.
- **Fully Functional Dual-API System:** With real-time feedback loop for continuous improvement.

Success

Successfully converted a semi-automatic machine into an intelligent, adaptive washing system.

Conclusion & Future Scope

Conclusion:

- Demonstrated a low-cost, AI-integrated laundry automation system.
- Bridges computer vision, prompt engineering, embedded systems, and mobile development.

Future Scope:

- Add weight sensor for load-based optimization.
- Cloud dashboard for wash history analytics.
- Support for multi-cloth batch scanning.
- Integration with smart home ecosystems (Google Home, Alexa).

Thank You!

Questions?

We welcome your feedback.